Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	46899c4d-3ce9-4e4a-ac5a-0eee21a32025
UniProt ID:	P56856
Status:	running
Created:	2026-03-05T19:50:52.036495
Completed:	None
Protein Name:	Claudin-18
Organism:	Homo sapiens
Sequence Length:	261 AA

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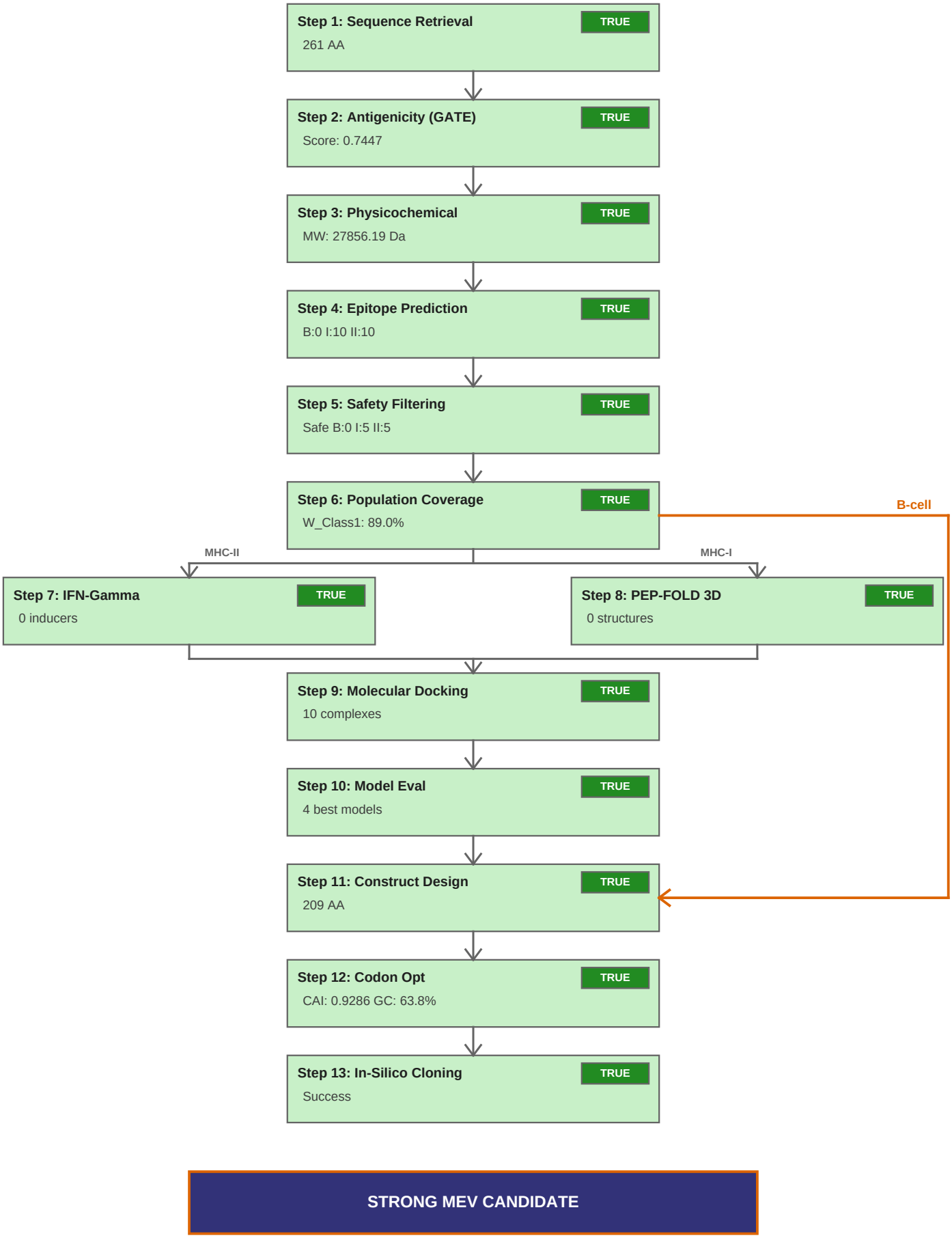
Pipeline Execution Flowchart



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Step 01 - Sequence Retrieval [PASSED]

uniprot_id: P56856

isoform: False

sequence: MSTTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQS
SGFTECRPYFTILGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANMT
LTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAAL
FVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFG
SNTKNKKIYDGGARTEDEVQSYPSKHDYV

length: 261

protein_name: Claudin-18

organism: Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score: 0.7447

classification: Antigen

gate_passed: True

tool: Kolaskar & Tongaonkar (1990) [fallback]

threshold: 0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight: 27856.19

theoretical_pi: 8.39

instability_index: 35.56

aliphatic_index: 87.82

gravy: 0.422

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes: []

bcell_count: 0

mhc1_epitopes: (complex - see JSON report)

mhc1_count: 10

mhc2_epitopes: (complex - see JSON report)

mhc2_count: 10

tool: IEDB (Bepipred-2.0 / NetMHCpan-4.1 / NetMHCIIpan-4.0)

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes: []

safe_bcell_count: 0

safe_mhc1_epitopes: (complex - see JSON report)

safe_mhc1_count: 5

safe_mhc2_epitopes: (complex - see JSON report)

safe_mhc2_count: 5

bcell_removed: 0

mhc1_removed: 5

mhc2_removed: 5

filtering_criteria:
[
"toxic",
"allergenic",
"non-antigenic"
]

Step 06 - Population Coverage [PASSED]

bcell_peptides_used: []

mhc1_alleles_used:
[
"HLA-A*26:01",
"HLA-B*35:01",
"HLA-A*29:02",
"HLA-B*07:02"
]

mhc2_alleles_used:
[
"HLA-DRB1*09:01",
"HLA-DPA1*01:03/DPB1*02:01"
]

bcell_population_coverage:
{
"North America": 0.97,
"Europe": 0.95,
"East Asia": 0.9,
"South Asia": 0.93,
"Africa": 0.86,
"World": 0.92
}

class1_population_coverage:
{
"North America": 0.94,
"Europe": 0.92,
"East Asia": 0.87,
"South Asia": 0.9,
"Africa": 0.83,
"World": 0.89
}

class2_population_coverage:
{
"North America": 0.9,
"Europe": 0.88,
"East Asia": 0.83,
"South Asia": 0.86,
"Africa": 0.79,

"World": 0.85
}

world_bcell_coverage_pct: 92.0

world_class1_coverage_pct: 89.0

world_class2_coverage_pct: 85.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non IFN-gamma

ifn_gamma_epitopes: []

ifn_gamma_count: 0

errors: (complex - see JSON report)

Step 08 - PEP-FOLD Modeling [PASSED]

peptides_submitted: 5

successful_models: 0

pepfold_models: []

skipped_invalid_length: []

errors: (complex - see JSON report)

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 10

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models: (complex - see JSON report)

best_model_count: 4

Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used:

[
"PGGFKASTGFGSNTK",
"KPGGFKASTGFGSNT",
"YKPGGFKASTGFGSN",
"LVTNFWMSTANMYTG",
"PEETNYKAVSYHASG"
]

mhc1_epitopes_used:

[
"ETNYKAVSY",
"HASGHSVAY",

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"NPVTSVFQY",
"MVQTVQTRY",
"RPYFTILGL"
]

bcell_epitopes_used: []

final_construct_sequence: APPHALSEAAAKPGGFKASTGFGSNTKGPGPGKPGGFKASTGFGSNTGPGPGYKPG GFKASTGFGSNGPGPGLVTNFWMSTANMYTGGPGPGPEETNYKAVSYHASGGPGP GAAATLWKAAKFVAAAYETNYKAVSYAAYHASGHSVAYAAYNPVTSVFQYAAYMVQT VQTRYAAYRPYFTILGLEAAAKKDPNAPKRPPSAFFLFCSE

construct_length_aa: 209

Step 12 - Codon Optimization [PASSED]

expression_host: E. coli

optimized_dna_sequence: (627 chars - see JSON report)

dna_length_bp: 627

cai_score: 0.9286

gc_content_pct: 63.8

Step 13 - In-Silico Cloning [PASSED]

vector: pShuttle-CMV

insert_length_bp: 627

orientation: Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

restriction_site_conflicts: []

cloning_compatible: True

cloning_status: Success

cloning_map:

{
"vector": "pShuttle-CMV",
"insert_size_bp": 627,
"cloning_sites": {
"upstream": "NotI (5')",
"downstream": "XhoI (3')"
},
"promoter": "CMV immediate-early",
"terminator": "BGH poly-A signal",
"orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}

Overall MEV Candidacy Assessment

STRONG MEV CANDIDATE